

GET POST API

TYPES OF APIs

EXPLAINED:

COMPLETE BEGINNER

TO ADVANCED GUIDE



OPEN API



PARTNER API



REST API



PRIVATE API



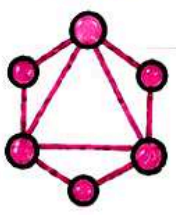
WEBHOOK API



INTERNAL API

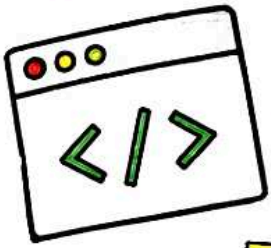


GraphQL API

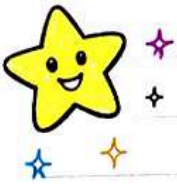


SOAP API

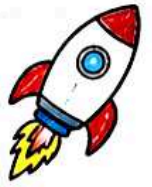
REST API



{...}



API



(Application Programming Interface)

An **API** allows two software applications to **communicate** with each other.

API

★ 1. Real-Life Example



Restaurant Example:

You = Customer
Waiter = API
Kitchen = Server
Food = Data



You place an order through the **waiter (API)**, and the kitchen sends back the **food (response)**.

★ 2. Why APIs Matter

- ✓ Connect applications
- ✓ Share data securely
- ✓ Save development time
- ✓ Enable automation
- ✓ Power modern web and mobile apps



3. Common API Use Cases

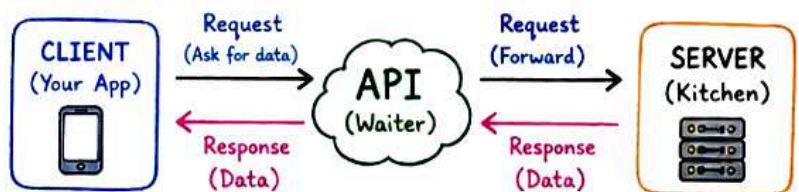
- Login with Google
- Weather apps
- Payment gateways
- Maps integration
- Social media sharing



4. How API Works? (Step-by-Step)

- 1 Client (App) makes a request to the API.
- 2 API receives the request and sends it to the Server.
- 3 Server processes the request and gets the data.
- 4 Server sends the data back to the API.
- 5 API sends the response back to the Client (App).

5. Simple Flow Diagram



6. Common API Methods (HTTP Methods)

Method	Use	Example
GET	Get data from server	GET /users
POST	Send new data to server	POST /users
PUT	Update existing data	PUT /users/1
DELETE	Delete data	DELETE /users/1

HTTP is the language APIs use to talk!



★ TIP

APIs are like messengers that help apps talk, share data, and work together!



7. Quick Summary

- ★ API = Bridge between applications.
- ★ It allows **request** → **process** → **response**.
- ★ Used everywhere: Web, Mobile, Cloud, AI, etc.
- ★ Makes our digital life **smart** and **connected**!

APIs are **Awesome!**



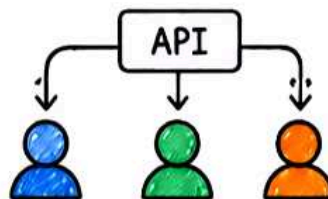


INTERNAL APIS (PRIVATE APIS)



1. WHAT ARE INTERNAL APIS?

Internal APIs are **used only** within an **organization**.



Internal APIs help different parts of a company talk to each other securely! 😊

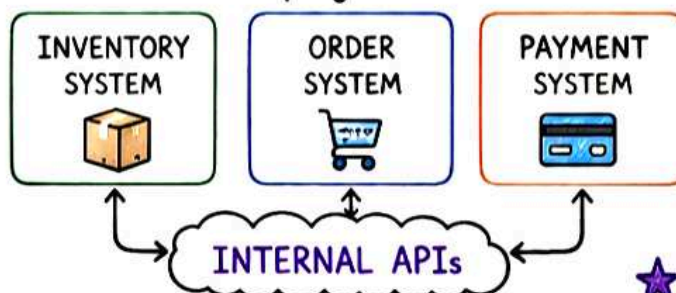
2. FEATURES

- ✓ Not **publicly** accessible
- ✓ Used by **company teams**
- ✓ Improved **security**
- ✓ Better **control**



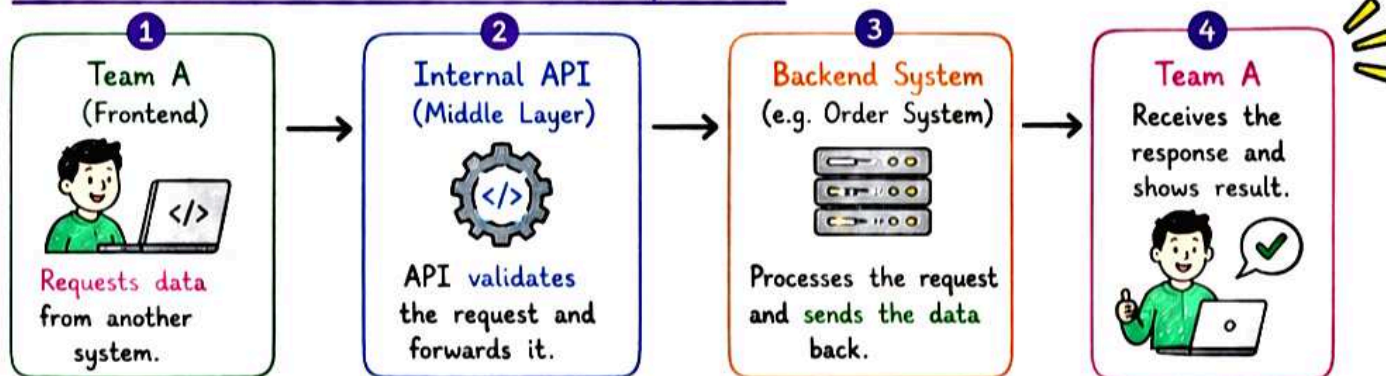
3. EXAMPLE

E-commerce company:



All these systems **communicate** with each other using **Internal APIs**.

4. HOW INTERNAL APIS WORK? (Simple Flow)



Internal APIs act like **messengers** inside the company to **connect** different systems.



5. BENEFITS

- Increased **productivity**
Teams can work faster with reusable APIs.
- Better **system integration**
Different systems work smoothly together.
- Enhanced **security**
Data stays within the organization.

6. USE CASES

- Enterprise software**
Connect different modules and departments.
- Banking systems**
Secure communication between services.
- Healthcare systems**
Link patient data, appointments, billing, etc.
- Internal dashboards**
Fetch data from multiple internal sources.

TIP

- ★ Internal APIs are **not** for the public.
- ★ They are powerful tools that help companies build **secure**, **scalable** and **efficient** systems!



APIs make teamwork possible!



OPEN APIs (PUBLIC APIs)



1. WHAT ARE OPEN APIs?

Open APIs are **publicly available** APIs that anyone can use.



2. FEATURES

- ✓ **Public** access
- ✓ **Easy** integration
- ✓ Usually require **API keys**
- ✓ Developer-friendly **documentation**



3. EXAMPLES

- ★ Google Maps API
- ★ OpenWeather API
- ★ GitHub API
- ★ News API



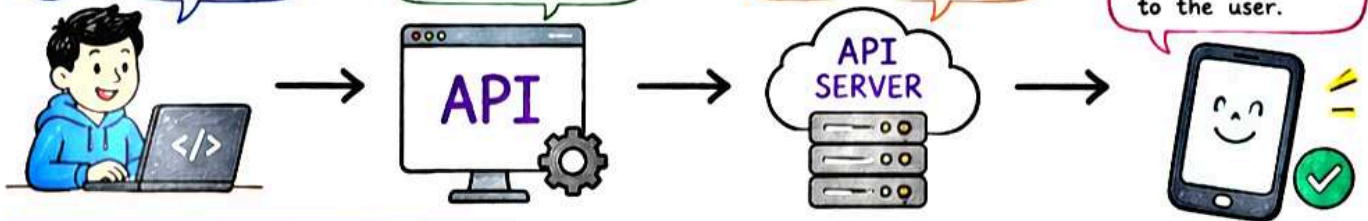
HOW OPEN APIs WORK? (Simple Flow)

1. You (Developer) build an app and need some data.

2. Your App sends a request to the Open API.

3. API Server processes the request and sends data back.

4. Your App receives the data and shows it to the user.



Example: Your weather app → asks OpenWeather API → gets weather data → shows to user.

4. ADVANTAGES

- ✓ **Fast** development
- ✓ Large developer **community**
- ✓ Rich **documentation**
- ✓ Ready to use **features & data**



5. DISADVANTAGES

- ✗ **Rate limits**
- ✗ **Security concerns**
- ✗ Limited **customization**
- ✗ Dependence on **third-party service**



6. BEST FOR

- **Startups**
- **Students**
- **Developers** building projects



✦ EXAMPLE USE CASES ✦



Map & Location
(Google Maps API)



Weather Info
(OpenWeather API)



Code & Repos
(GitHub API)



News & Articles
(News API)

QUICK TIP

Always read the API docs and follow their usage limits. Use **API keys** securely!



★ Open APIs make the web **connected**, **easy** to build, and **powerful**! ★





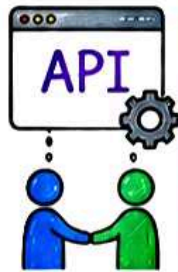
PARTNER APIs



APIs shared with **trusted business partners**

1. WHAT ARE PARTNER APIs?

Partner APIs are shared with **specific business partners** to enable collaboration and create value together.



In Simple Words

It's like giving a key to a trusted partner so they can access certain rooms, not the whole house!

2. FEATURES

- ✓ **Restricted** access
- ✓ Requires **agreements**
- ✓ **Secure** communication
- ✓ **Controlled** usage



3. EXAMPLE - TRAVEL WEBSITE

A travel website works with partners using Partner APIs.



API

HOTEL PARTNER API

Get hotel info, availability & booking



API

FLIGHT BOOKING API

Search flights, prices & booking



API

PAYMENT PROVIDER API

Process payments securely



4. HOW PARTNER APIs WORK? (STEP-BY-STEP)

1

Partner and company sign an agreement.



2

Partner gets API access (keys/credentials).



3

Partner makes API requests as per rules.



4

Your system processes the request.



5

Partner gets the response / data.



5. BENEFITS



Business collaboration

Work together to deliver better services.



Revenue opportunities

Monetize APIs through trusted partners.



Controlled ecosystem

You decide who gets access and how they use it.

6. INDUSTRIES USING PARTNER APIs



TRAVEL

Hotel bookings, flight search, itinerary services



FINANCE

Bank data sharing, payments, loan services



E-COMMERCE

Product data, inventory, order tracking



LOGISTICS

Shipment tracking, delivery partner integration

QUICK TIP

Always use Partner APIs securely and follow the agreed rules.

7. QUICK SUMMARY

- ☆ Partner APIs = Shared with specific business partners.
- ☆ They are secure, controlled and based on agreements.
- ☆ Helps in collaboration, growth and new revenue.
- ☆ Used in many industries to connect systems.

Stronger Partnerships
Stronger Business!





SOAP API

(Simple Object Access Protocol)



1. WHAT IS SOAP?

SOAP (Simple Object Access Protocol) is a protocol for exchanging structured information in web services.



In Simple Words

SOAP is like a formal letter with a strict format (XML) that ensures messages are delivered securely and correctly. 😊



2. FEATURES

✓ XML-based



✓ High security



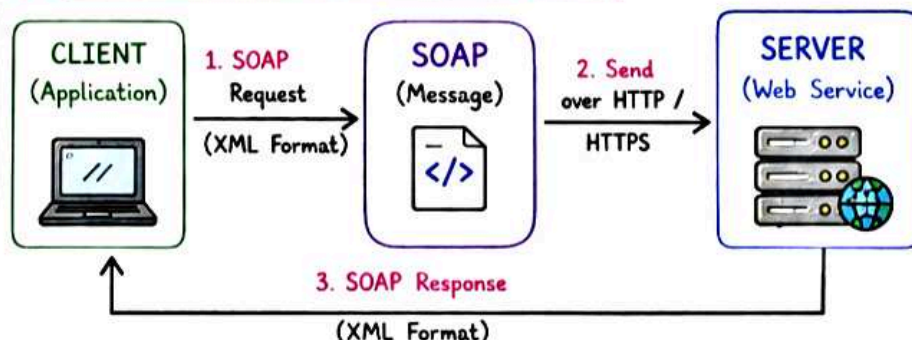
✓ Reliable messaging



✓ Enterprise-grade



3. HOW SOAP API WORKS (FLOW)



Example SOAP Request (XML)

```
<soap:Envelope>
  <soap:Header> ... </soap:Header>
  <soap:Body>
    <GetUserRequest>
      <UserID>101</UserID>
    </GetUserRequest>
  </soap:Body>
</soap:Envelope>
```

Key Point

SOAP uses XML for both request and response, making it platform independent. ★

4. ADVANTAGES

✓ Strong security



✓ Standardized communication



✓ Transaction support



✓ Reliable and error handling is built-in



5. DISADVANTAGES

✗ Complex



✗ Slower than REST



✗ Large message size



✗ Harder to learn



6. COMMON INDUSTRIES



Banking



Healthcare



Government systems



Enterprise software

7. SOAP vs REST (Quick Comparison)

Feature	SOAP API	REST API
Protocol	Uses XML over HTTP/HTTPS	Uses JSON / XML over HTTP
Security	Built-in (WS-Security)	Depends on implementation
Performance	Slower	Faster
Ease of Use	Complex	Simple
Best For	Enterprise, Critical Systems	Web & Mobile Applications

QUICK TIP

Use SOAP when you need high security, reliability and working with enterprise systems. Use REST for speed and simplicity! ★



SOAP API - Secure, Reliable, Standardized!





REST API

(Most Popular API)



1. WHAT IS REST API?

REST (Representational State Transfer) is the **most commonly** used API architecture.

REST API

In Simple Words

REST API is a set of rules that allows two applications to talk to each other over the internet using HTTP.

2. USES HTTP METHODS

METHOD	PURPOSE	DESCRIPTION	EXAMPLE	
GET	Retrieve Data	Used to get data from the server.	GET /users	
POST	Create Data	Used to send data to the server and create a new resource.	POST /users	
PUT	Update Data	Used to update an existing resource on the server.	PUT /users/1	
DELETE	Remove Data	Used to delete a resource from the server.	DELETE /users/1	

3. EXAMPLE ENDPOINT

<https://api.example.com/users>

Protocol
(Secure)

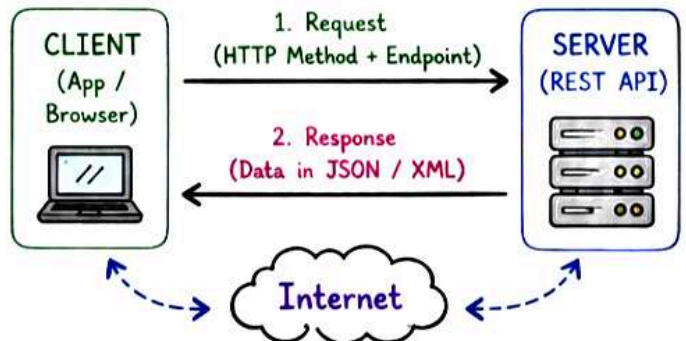
Domain
(API Server)

Resource
(What we want)

Real Example:

- GET <https://api.example.com/users> → Get all users
- POST <https://api.example.com/users> → Create a user
- PUT <https://api.example.com/users/1> → Update user
- DELETE <https://api.example.com/users/1> → Delete user

4. HOW REST API WORKS? (FLOW)



★ REST APIs usually return data in JSON format.



5. ADVANTAGES

- ✓ Simple
Easy to understand and use.
- ✓ Fast
Uses HTTP, so it's very efficient.
- ✓ Scalable
Works well for small and large applications.
- ✓ Easy to learn
Beginners can start quickly.



6. COMMON USAGE



- Web applications**
Connect front-end with back-end services.
- Mobile apps**
Fetch data and send requests from mobile.
- SaaS platforms**
Integrate with different third-party services.
- Cloud services**
Manage and control cloud resources.



QUICK TIP

7. QUICK SUMMARY

- ★ REST API is the most popular API style.
- ★ Uses HTTP methods like GET, POST, PUT, DELETE.
- ★ Simple, fast, scalable and easy to learn.
- ★ Used everywhere in web, mobile and cloud apps!

REST API makes the web more connected!



GRAPHQL API

API

Powerful • Flexible • Efficient

1. WHAT IS GRAPHQL?

GraphQL is a **query language** for APIs developed by **Meta**.



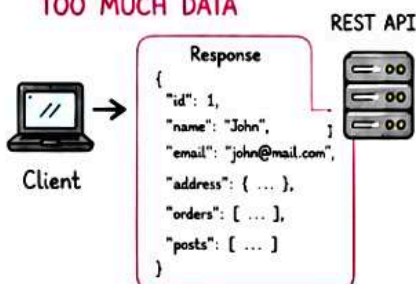
In Simple Words

GraphQL lets the client ask for exactly the data it needs, nothing more, nothing less! 😊

2. TRADITIONAL REST PROBLEM

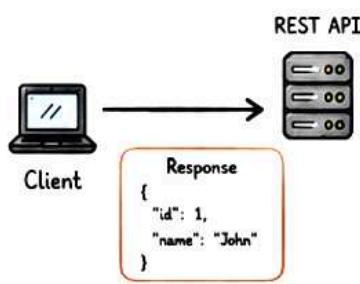
REST may return:

TOO MUCH DATA



Client gets more data than it actually needs. 😞

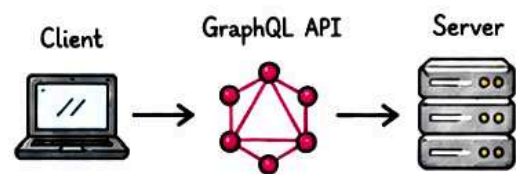
TOO LITTLE DATA



Client needs more data so it must make multiple requests. 😞

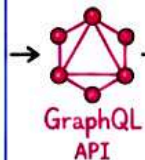
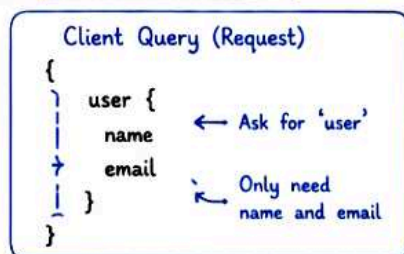
3. GRAPHQL SOLUTION

Request **exactly** what you need.



Client asks only for the data it needs and gets a precise response. 😊

4. EXAMPLE QUERY



Only the requested fields are returned! ✓

Key Point

GraphQL improves performance and developer experience! ⭐

5. BENEFITS

- ✓ Faster data fetching
- ✓ Reduced bandwidth
- ✓ Flexible queries
- ✓ Better frontend performance



6. USED BY

- ∞ Meta (Facebook, Instagram, WhatsApp)
- 🛒 Shopify
- 🐙 GitHub
- ☁ Modern SaaS products

Fun Fact!

GraphQL was created by Facebook in 2012 and open sourced in 2015. 😊

7. GRAPHQL vs REST (QUICK COMPARISON)

Feature	REST API	GraphQL API
Data Fetching	Multiple endpoints, may return too much or too little data	Single endpoint, get exactly what you need
Performance	Slower (multiple requests)	Faster (single request)
Bandwidth	Higher	Lower
Flexibility	Limited	Highly flexible
Best For	Simple applications	Complex, dynamic applications

QUICK TIP

Use GraphQL when your app has many data needs and you want better control over what you get! ❤️

Think of GraphQL as a smart waiter who brings you exactly what you order!



GraphQL = Ask less, Get more, Build better! 🎯

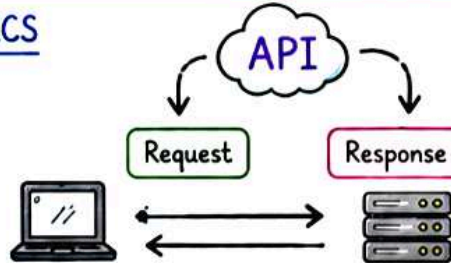
API LEARNING ROADMAP

(Beginner to Advanced)

1

STEP 1: LEARN BASICS

- ✓ What is API
- ✓ Request & Response
- ✓ HTTP Methods
- ✓ Status Codes



Notes

APIs allow different applications to talk to each other and share data.

2

STEP 2: LEARN REST APIS

- ✓ GET, POST, PUT, DELETE
- ✓ JSON Data
- ✓ API Testing

HTTP METHODS

GET	Retrieve data
POST	Create data
PUT	Update data
DELETE	Remove data

TOOLS

- Postman
- Insomnia



POSTMAN



INSOMNIA

3

STEP 3: AUTHENTICATION

- ✓ API Keys
- ✓ JWT Tokens
- ✓ OAuth 2.0



API KEY



OAuth 2.0

Tips

Practice regularly and build projects. That's the best way to master APIs!

4

STEP 4: ADVANCED APIS

- ✓ GraphQL
- ✓ SOAP
- ✓ WebSockets
- ✓ Rate Limiting



GraphQL



SOAP



WebSockets



Rate Limiting

5

STEP 5: BUILD PROJECTS

BEGINNER

- Weather App
- Quote Generator

INTERMEDIATE

- Blog API
- Task Manager

ADVANCED

- E-commerce Backend
- Chat Application
- SaaS Platform API

FINAL GOAL



LEARN APIS



BUILD PROJECTS



CREATE PORTFOLIO



GET INTERNSHIPS & JOBS



Keep Learning. Keep Building. Your Future is in Your Hands!